

## Xorstanta (xorstanta)

Giorgio is a big fan of board games and he has asked William to play with him. This time they are going to play a cooperative game, which means they need to play together in order to maximize their score. They have a lot of free time, so they are going to play  $T$  matches.



Figure 1: The numbers used in the game.

The rules of the game are very simple. First of all, in the  $i$ -th match they choose a number  $N_i$ . Then, for each number from 1 to  $N_i$  they need to decide whether Giorgio or William keeps it. Let's call  $A$  the set of numbers given to Giorgio and  $B$  the set of numbers given to William. Let  $X_A$  be the *bitwise xor* of all the numbers in  $A$  and  $X_B$  the *bitwise xor* of all the numbers in  $B$ . The score of the match is given by the sum of  $X_A$  and  $X_B$ . What is the maximum score they can achieve in each match?

 Among the attachments of this task you may find a template file `xorstanta.*` with a sample incomplete implementation.

### Input

The first line contains the only integer  $T$ , the number of matches. Each one of the next  $T$  lines contains one integer  $N_i$ , the number chosen in the  $i$ -th match.

### Output

For each match, you need to write a single line with an integer: the maximum score that can be achieved.

## Constraints

- $1 \leq T \leq 100\,000$ .
- $1 \leq N_i \leq 10^9$  for each  $i = 0 \dots T - 1$ .

## Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points)      Examples.  

- **Subtask 2** (10 points)       $T = 1$  and  $N_0 \leq 20$ .  

- **Subtask 3** (20 points)       $T = 1$  and  $N_0 \leq 100\,000$ .  

- **Subtask 4** (70 points)      No additional limitations.  


## Examples

| input        | output  |
|--------------|---------|
| 2<br>6<br>15 | 7<br>30 |

## Explanation

In the **first sample case** there are two matches.

In the first match,  $N = 6$  and the best score is 7, which can be achieved by giving all the numbers from one to six to Giorgio. In this case  $A = \{1, 2, 3, 4, 5, 6\}$ ,  $B$  is empty,  $X_A = 7$  and  $X_B = 0$ .

In the second match,  $N = 15$  and the best score is 30, which can be achieved by giving the numbers 1, 5 and 11 to Giorgio and the rest to William. In this case  $A = \{1, 5, 11\}$ ,  $B = \{2, 3, 4, 6, 7, 8, 9, 10, 12, 13, 14, 15\}$ ,  $X_A = 15$  and  $X_B = 15$ .